Harish Vijayakumar

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Education

Georgia Institute of Technology Atlanta, GA

Bachelor of Science in Computer Engineering, GPA: 3.73/4.0 *Expected May 2028*

Relevant Coursework: Physical Foundations of Computer Engineering (Current), Programming for Hardware/Software Systems, Digital System Design, Circuit Analysis, Data Structures and Algorithms

Experience

RevSync January 2026 – Present

Lead Hardware Engineer *Atlanta, GA*

- Contributed to initial design and hardware requirements while testing sensing, compute, and vehicle interfaces.
- Co-presented RevSync at the GT Klaus Startup Challenge, communicating technical direction and roadmap, achieving acceptance to CREATE-X LAUNCH.

Cultivate Lab January 2025 – January 2026

Undergraduate Researcher *Athens, GA*

- Co-authored 2 engineering education publications examining college student mental health and coping strategies.
- Screened 5,000+ peer-reviewed articles using Rayyan to extract and synthesize data from 35 qualifying studies.

FPGA UART Communication & DSP Platform | *SystemVerilog, DE10-Lite* May 2026 – June 2026

- Designed and implemented a modular SystemVerilog UART communication platform featuring UART TX/RX, FIFO buffering, command parsing, and FSM controller, supporting 115200 baud and meeting 50 MHz timing.
- Developed a DSP subsystem including a moving average filter, modifiable clock divider, and 7-segment display driver; utilized 503 logic elements and 282 registers, achieving 125 MHz Fmax and +12.0 ns setup slack.
- Verified RTL functionality using ModelSim and UART with CP2102 USB-to-UART interface and Arduino IDE.

8-Bit CPU | *SystemVerilog, ModelSim, DE10-Lite* June 2026 – July 2026

- Created an 8-bit processor datapath implementing a program counter, instruction memory/decoder, register file, arithmetic logic unit (ALU), and FSM controller with 4 general-purpose registers meeting 50 MHz timing.
- Implemented five logic instructions (ADD, SUB, AND, OR, XOR) with register writeback and instruction execution.
- Verified RTL components using ModelSim and DE10-Lite hardware; utilized 70 registers and 150 logic elements, achieving 105 MHz Fmax, +10.5 ns setup slack, and 0 TNS.

ASIC Calculator Chip Design - SiliconJackets | *Xcelium, SimVision, Innovus* January 2026 – March 2026

- Implemented a SystemVerilog 64-bit SRAM-based arithmetic calculator with SRAM-based operand storage and FSM, combining upper and lower 32-bit operations with carry propagation.
- Verified the design using Xcelium, SimVision, and Innovus with a driver, monitor, sequencer, scoreboard, 3 assertions, and 400 constrained-random transactions, achieving 99.95% coverage and 100% assertion coverage.
- Resolved physical design rule check (DRC) violations in Cadence Innovus by reordering metal routing layers, achieving a DRC-clean final layout.

Publications

Survey Instruments Used to Study Mental Health Among STEM Graduate Students
IEEE Frontiers in Education Conference (FIE), 2025

Photoelicitation and Coping Strategies Among First-Generation and Low-SES Engineering Students
ASEE Annual Conference & Exposition, 2026

Technical Skills

Languages: SystemVerilog, Verilog, VHDL, C/C++, Java, RISC-V Assembly, Python, MATLAB
Digital Design: RTL Design, Finite State Machines, Computer Architecture, FPGA Design
Verification: ModelSim, Cadence Xcelium, RTL Simulations, Waveform Debugging
Tools: Intel Quartus Prime, Cadence Innovus, Git, Linux
Hardware: Intel DE10-Lite FPGA, Arduino, Oscilloscopes, Logic Analyzer
Certifications: OSHA, CITI Human Subject Research
Organizations: SiliconJackets, GT Boxing Club (Secretary), Beta Chi Theta National Fraternity